a Scientific input & delivery DAG

Research intent

Molecules · reaction conditions
Competing pathways & methods
Natural-language hypothesis formation

Delivery DAG

R —TS→ P

Species nodes · reaction-step edges

Default: ts_opt · frequency · short_irc

Step query

b Geometry & PES

Atom mapping

Atom-conserving candidates
Reaction-center atom pairs

Steric traversal

Distance-weighted local bulk
Substituent contributions, ρ^d

Rigid alignment

Kabsch endpoint alignment
Consistent endpoint frames

Path preparation

NEB–IDPP · light IDPP
Candidate paths & TS seeds

Evidence binding

Guided diagnostics

Cloudra

Cloudra Agent System

Job dispatch

d Mechanistic validation & refinement

Identity-resolved pathways

TS
R P

Same TS → ±IRC → Opt/Freq
Atom mapping · species identity
n_imag(TS) = 1 · n_imag(R/P) = 0

Reaction-centered modes

Pair projections · atomic motion
Reaction-center contributions

Evidence-guided revision

Locate conflicting evidence
Revise affected calculations

Native outputs

c Computational infrastructure

Scheduler

Per-node quotas · Pinned hosts
Independent of the web API

ComputeDB ledger

Input hashes · Deduplication
Durable records · recovery

Engine execution

ORCA · Gaussian · xTB
Native logs · file locks
Geometries · energies · frequencies